Programme Name/s	**: Mining & Mine Surveying/ Mining Engineering**
Programme Code	**: MS/ MZ**
Year	**: Second**
Course Title	**: MINE VENTILATION**
Course Code	**: 332304**

I. RATIONALE

The underground working is devoid of the natural air. As such to make the working places safe for the persons to work and pass it is necessary to circulate the air artificially through the mine working. A mining engineer should be aware of the principles of how the flow of air can be created, regulated, controlled and monitored. They should be aware of the heat effect and humidity, condition and means of measuring and controlling the same. Number of mine gases is produced in the mine, which has got dangerous and toxic properties.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

Aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences: • Evaluate presence of inflammable and toxic/noxious gases in the general body of air and undertake the ventilation survey as per requirement.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Use various instruments for detection of inflammable and toxic/noxious gases in the mine.
- CO2 - Measure hunnidity and cooling power of the mine air.
- CO3 - Identify the ventilation system of a mine.
- CO4 - Identify various ventilation appliances used in mine.

- CO5 - Use various instruments in pressure and quantity survey.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											Total Marks
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
											Max	Max	Max	Min	Max	Min	Max	Min	Max	Min	
332304	MINE VENTILATION	MVE	DSC	3	-	2	-	5	5	3	30	70	100	40	25	10	25#	10	-	-	150

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- ClassRoom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @$ Internal Online Examination

Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 30 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the method of detection of given damps. TLO 1.2 Explain the given tests carried out by Flame Safety Lamp. TLO 1.3 Describe Methanometer and measurement of methane using Methanometer. TLO 1.4 Explain principle of Gas Chromatography.	**Unit - I Mine Air** 1.1 Different Gases / Damps found in mines, Definition of damps, their threshold limits, physiological effects, source of production and detection, Degree of gassiness of seam, multi gas detector. 1.2 Flame safety lamps, its principle, construction, safety features, and comparison. Detection of Methane by flame safety lamp. 1.3 Methanometer its principle of working, construction. Principle of following methods of detection of methane. a)Formation of gas cap of Flame safety lamp b)Wheatstone bridge principle c)Diffusion-Combustion - Contraction d)Difference in refractive index of pure air and methane e)Infrared spectrometry f)Reaction of methane with chemicals. 1.4 Principle of chromatography and types of gas chromatography with its applications.	Lecture Using Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on Project based

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Explain the objectives and standards of ventilation. TLO 2.2 Describe the method of calculation of ventilating pressure. TLO 2.3 Calculate relative humidity of mine air under given condition. TLO 2.4 Interpret effect of heat and humidity on miners. TLO 2.5 Calculate the motive column & Natural Ventilation Pressure (NVP).	**Unit - II Mine Climate** 2.1 Purpose and standards of ventilation, standards for minimum & maximum velocity of air for different locations. 2.2 Measurement of air Pressure, Water gauge, total pressure, Measurement of Ventilating pressure using an inclined Manometer. 2.3 Sources of heat and humidity in mines, Moisture content of mine air, relative humidity, wet bulb temperature, measurement and calculation of relative humidity using Whirling Hygrometer. 2.4 Cooling power of mine air, determination of cooling power by using Kata Thermometer, methods of improving cooling power of mine air, effect of heat and humidity on miners. 2.5 Natural ventilation Pressure, geothermic gradient, Factors causing NVP, Motive column, calculation of natural ventilation pressure.	Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on
3	TLO 3.1 Compare given types of fans used for mine ventilation. TLO 3.2 Calculate different efficiencies and theoretical depression produced by fan.	**Unit - III Artificial Ventilation** 3.1 Different types of fans used in mines: centrifugal & axial flow, their principle of working, Exhaust & forcing type. Purposes of evasee & volute casing. Fans in series and parallel, Comparison between axial flow & Centrifugal fan; exhaust & forcing Fan. 3.2 Fan laws, Manometric efficiency overall efficiency, theoretical depression produced by fan. Calculation of water guage, BHP, efficiency, theoretical depression using fan laws.	Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Calculate the quantity of air flowing in mines under given condition. TLO 4.2 Describe the given ventilation devices & auxiliary fans. TLO 4.3 Describe the given ventilation devices& auxiliary fans.	**Unit - IV Distribution & Coursing of Air in Mines** 4.1 Laws of air flow in Mines, Atkinson's formula, splitting, its advantages & disadvantages, Calculation of quantity of air in each split, Equivalent orifice: Definition and formula. 4.2 Ventilation appliances, Auxiliary ventilation: Different methods, advantages & disadvantages, hazards associated with auxiliary ventilation, precautions required. Booster fan: purpose, dangers associated, precautions before installation, numerical on booster fan. 4.3 Ascensional and Descensional ventilation, advantages and disadvantages	Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on
5	TLO 5.1 Describe given type of ventilation survey. TLO 5.2 Describe given instruments to measure the quantity and pressure of mine air. TLO 5.3 Explain given ventilation survey.	**Unit - V Ventilation Survey** 5.1 Scope and importance of ventilation survey, survey interval and location of survey station, ventilation plan. 5.2 Measurement Of quantity & pressure difference, anemometer, pitot static tube. 5.3 Method of conducting Pressure & quantity survey, precautions during and before conducting ventilation survey, conventional signs used in ventilation survey.	Chalk-Board Demonstration Video Demonstrations Site/Industry Visit Hands-on

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Detect percentage of Firedamp/Whitedamp/Blackdamp/Stinkdamp/oxygen.	1	*Use of Multigas detector (Drager Xam 5600/Mini Warn)	2	CO1
LLO 2.1 Identify various parts of MSA Methanometer and detecting percentage of methane.	2	*Use of MSA Methanometer.	2	CO1
LLO 3.1 Identify GL5, GL50, GL60 & GL7 flame safety lamps	3	*Flame safety lamps.	2	CO1
LLO 4.1 Dismantle & assemble of GL50 and GL7 Flame safety lamps.	4	*GL50 and GL7 Flame safety lamps.	2	CO1
LLO 5.1 Use flame safety lamp for accumulation test.	5	*Use of Flame safety lamp.	2	CO1
LLO 6.1 Use flame safety lamp for percentage test.	6	*Use of Flame safety lamp.	2	CO1
LLO 7.1 Identify various parts of whirling hygrometer.	7	*Whirling hygrometer.	2	CO2
LLO 8.1 Use whirling hygrometer to determine relative humidity of air.	8	*Use of whirling hygrometer.	2	CO2
LLO 9.1 Use kata thermometer to determine cooling power of mine air	9	*Use of Kata thermometer.	2	CO2
LLO 10.1 Identify various parts of centrifugal fan.	10	*Centrifugal mine fan.	2	CO3
LLO 11.1 Identify various parts of axial flow fan.	11	*Axial flow fan.	2	CO3
LLO 12.1 Identify various parts of air crossing.	12	*Air crossing model.	2	CO4
LLO 13.1 Identify various types of ventilation stopping.	13	*Ventilation stopping model.	2	CO4
LLO 14.1 Simulate working of a regulator.	14	*Regulator model.	2	CO4

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 15.1 Identify various parts of permanent stopping.	15	*Construction of Permanent stopping.	2	CO4
LLO 16.1 Identify various parts of an Air lock	16	Air lock model.	2	CO4
LLO 17.1 Use Vane Anemometer for calculating quantity of air as per given conditions.	17	*Use of Vane Anemometer.	2	CO5
LLO 18.1 Identify various parts of Velometer.	18	*Velometer.	2	CO5
LLO 19.1 Measure air velocity by Velometer.	19	*Use of Velometer	2	CO5
LLO 20.1 Identify various parts of inclined manometer.	20	Inclined manometer	2	CO5
LLO 21.1 Calculate velocity pressure by using Inclined Manometer.	21	Use of Inclined Manometer	2	CO5
LLO 22.1 Identify various parts of Pitot Static Tube.	22	Pitot Static Tube.	2	CO5
LLO 23.1 Calculate static pressure & velocity pressure by using Pitot Static Tube.	23	Use of Pitot Static Tube.	2	CO5
LLO 24.1 Draw various Conventional signs used in Ventilation Plan.	24	Conventional signs used in Ventilation Plan.	2	CO5

Note : Out of above suggestive LLOs -

- '*' Marked Practicals (LLOs) Are mandatory.
- Minimum 80% of above list of lab experiment are to be performed.
- Judicial mix of LLOs are to be performed to achieve desired outcomes.

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Prepare a report on measurement of different gases by using multigas detector by visiting nearby underground mine.
- Prepare a report on mine fan house regarding water gauge by visiting nearby underground mine.
- Prepare report on design parameters of evasee.
- Present a power point presentation on various types of fans used in Indian underground mines for ventilation purpose.
- Prepare any one model of ventilation devices.
- Conduct a ventilation survey in nearby underground mine and prepare a report of it.
- Prepare a report/paper on history of gas testing by old conventional methods.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicial mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Multigas detector(drager X-am 5600/ Mini warn:- It has an ergonomic design, and used for detection of exposive, combustible and toxic gases as well as oxygen.	1
2	Model of Centrifugal fan	10
3	Model of Axial flow fan.	11
4	Model of Air crossing.	12
5	Model of Ventilation stopping.	13
6	Model of Regulator.	14
7	Model of Airlock.	16
8	Vane Anemometer :- It is an instrument to determine the distance travelled by air in a given time and is used where the air velocities are between 60m/min and 1000m/min. Its possible measuring wind speed is 1-15m/sec.	17
9	Velometer :- It is an electronic device which directly measure the air velocity in m/sec at any point of observation. Its range varies from 0-15m/sec.	18,19
10	MSA Methanometer :- MSA D6 Methanometer is commonly used in our coal mines, It is suitable for spot checking of gas accumulation with measurement range from 0 to 5 % by volume, The battery lasts for nearly 1000 ten-sec and the complete charging takes about 14 hours.	2
11	Inclined Manometer :- Its is used for measuring pressure/ water gauge in mines. Maximum range of pressure measurement is 250mm Wg.	20,21
12	Pitot Static tube :- Its is used to measure to measure the static pressure, velocity pressure or the total pressure. L-type pitot tube feature a modified ellipsoidal head with a single forward facing hole (for total pressure) and a ring of side holes (for static pressure).	22,23

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
13	Flame safety lamp (GL5, GL50, GL60, GL7) :- It is used for detection of inflammable gases (Firedamp) in underground coal mines, It can detect accumulation and perecntage of firedamp in the atmosphere. It is manufactured by J.K. Dey & Sons, Kolkata.	3,4,5,6
14	Whirling hygrometer :- It is used to determine relative humidity in mine air. Its range generally varies from -5 to 50 degree celcius.	7,8
15	Kata thermometer :- It is used to measure the cooling power of mine air, It consist of an alcohol thermometer with a large bulb 4cm. long and 2cm. in diameter and a stern 20cm. long. It is graduated at two points namely 38 degree celsius and 35 degree celsius	9

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Mine Air	CO1	19	4	4	6	14
2	II	Mine Climate	CO2	19	4	4	6	14
3	III	Artificial Ventilation	CO3	20	2	4	10	16
4	IV	Distribution & Coursing of Air in Mines	CO4	18	2	4	10	16
5	V	Ventilation Survey	CO5	14	2	4	4	10
Grand Total				**90**	**14**	**20**	**36**	**70**

X. ASSESSMENT METHODOLOGIES/TOOLS

Formative assessment (Assessment for Learning)

- Progressive test
- Progressive test
- Lab performance
- Journal submission

Summative Assessment (Assessment of Learning)

- End Year Examination
- Viva-voce

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3			3	3		2			
CO2	3	2		3	3		2			
CO3	3	1	2	3			2			
CO4	3	1	2	3	3		2			

CO5	3	2	2	3	3	1	2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	D.J. Deshmukh	Elements of Mining Technology- Vol.2	Dennet, Nagpur, ISBN: 978-81-89904-34-0
2	G.B. Misra	Mine Environment and Ventilation	Oxford University Press, ISBN: 0 19 562232 4
3	L.C. Kaku	Numerical Problems on Mine Ventilation	Lovely Prakashan, Dhanbad, ISBN: 978-8179561393
4	L.C. Kaku	Gas Testing	Lovely Prakashan, Dhanbad, ISBN: 978-8179561522

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://youtu.be/Id4R-2KT9jc?si=xv-Z7yCrcHAewK2n	Use of Whirling Hygrometer to determine relative humidity of mine air.
2	https://youtu.be/tH3O9dlOvfs?si=B__hPrtzFfWqF47h	Types of fans used in mines for artificial ventilation.
3	https://youtu.be/JHb6fkqNyTY?si=Oda0_61_NTa_9cGh	Types of damps.
4	https://youtu.be/N7PSFm680Is?si=5X66qXmZZ9eRNlNW	Types of Flame Safety lamps used in mines.
5	https://youtu.be/Ww5BxnIkj_0?si=kJgOlGKHSC7niqkG	Concept of Water gauge and Ventilation pressure.
6	https://youtu.be/LmctN-Kw75Q?si=MG8upv0CjFl5Q90M	Instruments used in mines for ventilation survey.
7	https://youtu.be/6j6DZGcxocM?si=s59fsBvMGOhzrlsd	Concept of Methanometer for firedamp detection.

Sr.No	Link / Portal	Description

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students